

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403813
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Harris; Ryan C. et al.

Apparatus for preventing cargo movement in a vehicle

Abstract

Apparatuses described herein relate to improving the prevention of cargo movement during vehicle travel. In one embodiment, an apparatus for securing cargo within a vehicle includes a bladder coupled to an underside of a cover that extends over a cargo area, where the bladder faces the cargo area. The apparatus includes a supply device coupled with the bladder via a connecting member to expand the bladder into contact with the cargo within the cargo area.

Inventors: Harris; Ryan C. (Saline, MI), Frederick; Scott L. (Brighton, MI)

Applicant: Toyota Motor Engineering & Manufacturing North America, Inc. (Plano, TX)

Family ID: 87851871

Assignee: Toyota Motor Engineering & Manufacturing North America, Inc. (Plano, TX)

Appl. No.: 17/685963

Filed: March 03, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230278482 A1	Sep. 07, 2023

Publication Classification

Int. Cl.: B60P7/06 (20060101); B60P7/04 (20060101)

U.S. Cl.:

CPC B60P7/065 (20130101); B60P7/04 (20130101);

Field of Classification Search

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2907580	12/1958	Tietig	293/1	B60R 7/02
4792178	12/1987	Kokx	296/100.18	B60J 7/068
6537003	12/2002	Rostoker	410/125	B60P 7/065
7131805	12/2005	Morris	N/A	N/A
2002/0073629	12/2001	Finell	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2720057	12/1977	DE	B60P 7/065
102006057116	12/2006	DE	N/A
202013006707	12/2012	DE	N/A
1523421	12/2004	EP	N/A
3106332	12/2015	EP	N/A
WO-9601194	12/1995	WO	B60P 7/065
WO-2020053876	12/2019	WO	B60P 7/065

Primary Examiner: Weisberg; Amy R

Assistant Examiner: Adams; Philip Charles

Attorney, Agent or Firm: Darrow Mustafa PC

Background/Summary

TECHNICAL FIELD

(1) The subject matter described herein relates, in general, to preventing cargo movement within a vehicle, and, more particularly, to using an inflatable bladder that expands into contact with the cargo to prevent the cargo from moving.

BACKGROUND

(2) When cargo is placed in a cargo space of a vehicle, such as a truck bed or trunk, the cargo is free to move and slide as the vehicle travels. When cargo is free to move as the vehicle travels, cargo can break or become damaged. To prevent cargo from sliding and breaking during travel, drivers take inconvenient actions including taking slow turns, travelling at slow speeds, or holding the cargo as they drive. Moreover, existing mechanisms for securing cargo are cumbersome and include strapping cargo into place, placing cargo in pockets or nets, and placing cargo in designated compartments while also not necessarily securing cargo that is too large or small to fit into designated spaces. Further, these existing mechanisms require users to install equipment such as straps, pockets, nets, and compartment spaces within a cargo space, and additionally, these mechanisms require users to manually manipulate the equipment to secure the cargo.

SUMMARY

(3) Example apparatuses disclosed herein relate to improving the prevention of cargo movement during vehicle travel. As previously noted, when cargo is placed in a space, such as a truck bed or

trunk, cargo is free to slide and move, which can lead to cargo damage. As a result, drivers change driving maneuvers to prevent unsecured cargo from moving and breaking, install equipment, such as straps, pockets, nets, and compartments in cargo spaces to prevent cargo from moving, and manipulate equipment present in the vehicle to minimize cargo movement.

(4) Therefore, in one embodiment, a system is disclosed that improves the prevention of cargo movement in a vehicle by providing an apparatus for securing cargo as the vehicle travels. For example, in the context of a truck, a user may store the cargo in a truck bed, which is prone to cargo items sliding about. As such, in one configuration, the system includes a tonneau cover with a bladder coupled to the underside of the tonneau cover that extends over the truck bed. In one approach, the bladder is formed from a material that is resistant to puncture, such as rubber. This allows the bladder to come into direct contact with and to tightly secure cargo, including cargo with sharp edges and rigid shapes. A supply device, such as a compressor, fill valve, or pump is coupled with the bladder to expand the bladder into contact with the cargo within the cargo area by filling the bladder with water, pre-bottled condensed air, or compressed air, which prevents the cargo from moving. The system may be located in any area of the vehicle used to secure cargo, including a truck bed, a trunk space, and a seatback of the vehicle.

(5) The supply device may expand the bladder in response to an electronic input from a human machine interface (HMI) or from a switch located on the supply device. In one embodiment, the supply device includes a pressure sensor that senses the pressure within the bladder. The supply device, informed by the pressure sensor, expands the bladder to a threshold pressure to secure the cargo. In one approach, the bladder does not expand uniformly above the cargo space in order to conform around the various cargo or lack thereof in the cargo space. The threshold pressure may depend on the dimensions and weight of the cargo. As such, if the only cargo in the cargo space is a box, the threshold pressure informs the supply device to expand the bladder to touch and secure the box. The bladder will expand to the threshold pressure, which results in the bladder hugging the cargo as tightly as possible without puncturing the bladder and without deforming the cargo.

(6) The bladder remains expanded until, in one arrangement, a solenoid valve located on the supply device is actuated. When the solenoid valve actuates, the flow of air or water within the bladder reverses into the supply device, causing the bladder to compress.

(7) The bladder may comprise distinct zones. For example, in one configuration, the cover may be a trifold tonneau cover. As such, the bladder is separated into zones, each zone correlating to one section of the trifold tonneau cover. Accordingly, a pressure sensor is present to detect the pressure in each zone. The supply device expands each zone of the bladder separately to a threshold pressure for each zone. The threshold pressure may depend on the dimensions and weight of the cargo beneath the zone. As such, if the cargo in the cargo area is beneath the first zone of the bladder but not the second and third zones of the bladder, only the first zone of the bladder will expand to securely hug the cargo. As previously discussed, the first zone of the bladder remains expanded until a solenoid valve on the supply device is actuated to reverse the flow of air or water then-present in the bladder back into the supply device.

(8) In one embodiment, an apparatus for securing cargo within a vehicle is disclosed. The apparatus includes a bladder coupled to an underside of a cover that extends over a cargo area, where the bladder faces the cargo area. The apparatus includes a supply device coupled with the bladder via a connecting member to expand the bladder into contact with the cargo within the cargo area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate various systems, methods, and other embodiments of the disclosure. It will

be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one embodiment of the boundaries. In some embodiments, one element may be designed as multiple elements or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

(2) FIG. 1 illustrates one embodiment of an apparatus that is associated with improving the prevention of cargo movement in a cargo area.

(3) FIG. 2 illustrates one embodiment of the apparatus of FIG. 1 that is associated with improving the prevention of cargo movement in a cargo area by using a bladder that is expanded with a compressor.

(4) FIG. 3 illustrates one embodiment of an interior view of a cargo area implemented with the apparatus of FIG. 1.

(5) FIG. 4 illustrates one embodiment of the apparatus of FIG. 1 implemented with a trifold tonneau cover.

DETAILED DESCRIPTION

(6) Example apparatuses associated with improving the prevention of cargo movement are disclosed. Current methods for preventing cargo movement in a vehicle present difficulties for drivers. For example, when cargo is placed in a space, such as a truck bed or trunk, cargo is free to slide and move, which can lead to cargo damage. To prevent cargo damage, drivers must take actions, such as changing driving maneuvers to prevent unsecured cargo from moving and breaking, installing equipment, such as straps, pockets, nets, and compartments in cargo spaces to hold cargo tightly, and manipulating equipment present in the vehicle to minimize cargo movement (i.e., placing cargo in netting, strapping cargo down with restraints, installing compartments in a cargo area, etc.).

(7) Therefore, in one embodiment, an apparatus is disclosed that improves the prevention of cargo movement in a vehicle by securing cargo as the vehicle travels. In one embodiment, the apparatus includes a cover that is positioned above cargo in a cargo space of the vehicle, where the cover includes an inflatable bladder coupled to the underside of the cover. For example, in the context of a truck, a user may store the cargo in a truck bed. Accordingly, in one configuration, the cover is a tonneau cover with a bladder coupled to the underside of the tonneau cover extending over the truck bed. A supply device, such as a compressor, fill valve, or pump is coupled with the bladder to expand the bladder into contact with the cargo within the cargo area by filling the bladder with water, pre-bottled condensed air, or compressed air. The location of the supply device may be in any area with extra space in the vehicle, such as an inner deck panel of a truck bed, an underside of a vehicle seat, a trunk space, or the like. In one approach, the bladder is formed from a material that is resistant to puncture, such as rubber. This allows the bladder to come into direct contact with and to tightly secure cargo, including cargo with sharp edges and rigid shapes, without rupturing. The system may be located in any area of the vehicle used to secure cargo, including a truck bed, a trunk space, a seatback of the vehicle, and the like.

(8) The supply device may expand the bladder in response to an electronic input from a human machine interface (HMI) or from a switch located on the supply device. For example, the HMI may be a touchpad, such as a multimedia display within the vehicle, a button, or a switch located within the vehicle. In one approach, a driver interacts with the HMI to cause the supply device to expand the bladder. For example, a driver may touch a virtual button on the multimedia display to begin the flow of air or water coming from the supply device. In another configuration, a driver may activate the supply device manually by flipping a switch on the supply device.

(9) In one embodiment, the supply device includes a pressure sensor that senses the pressure within the bladder. The supply device, informed by the pressure sensor, expands the bladder to a threshold pressure to secure the cargo. In one approach, the bladder expands above the cargo space in a non-

uniform manner by conforming around the various cargo or within open space in the cargo area. In one configuration, the threshold pressure is manually input by a user on an HMI or directly on the supply device. In one embodiment, the user selects the threshold pressure from a list of preset threshold pressures.

(10) The threshold pressure may depend on the dimensions and weight of the cargo. In one arrangement, a user may input information on the supply device to establish an appropriate threshold pressure for expanding the bladder. In one approach, a user may input information on an HMI about the cargo, such as the dimensions and weight of the cargo. In one embodiment, the threshold pressure increases as the dimensions of the cargo decreases, and the threshold pressure increases as the weight of the cargo increases. In one arrangement, a user may input the exact dimensions and weight of the cargo on an HMI. Alternatively, a user may input generalized dimension and weight information about the cargo on an HMI. For example, if a user inputs that the cargo in the cargo space is small and light, a threshold pressure to secure the cargo will be determined for small and light cargo. The supply device will expand the bladder to the threshold pressure, which results in the bladder hugging the cargo as tightly as possible without puncturing the bladder and without deforming the cargo.

(11) The bladder remains expanded until, in one arrangement, the apparatus actuates a solenoid valve located on the supply device. The apparatus may actuate the solenoid valve in response to the supply device receiving an electronic input from a human machine interface (HMI) or in response to a user engaging with an input device, such as a switch located directly on the supply device. The HMI may be a touchpad, such as a multimedia display within the vehicle, a button, a switch, a door handle, a gear shifter, or a tailgate handle. For example, the supply device may begin compressing the expanded bladder when a user pulls on the tailgate of the vehicle. Alternatively, a user may instruct the supply device to compress the bladder manually by flipping a switch on the supply device. In any case, when the supply device receives an electronic input from an HMI or from a switch located on the supply device, a solenoid valve actuates. When the solenoid valve actuates, the flow of air or water within the bladder reverses into the supply device, causing the bladder to compress.

(12) The bladder may comprise distinct zones. For example, in one configuration, the cover may be a trifold tonneau cover. As such, the bladder is separated into zones, each zone correlating to one section of the trifold tonneau cover. Accordingly, a pressure sensor corresponds to each zone. The supply device expands each zone of the bladder separately to a threshold pressure for each zone. The threshold pressure may depend on the dimensions and weight of the cargo beneath the zone, which may be input by a user. As such, if the cargo in the cargo area is beneath the first zone of the bladder but not the second and third zones of the bladder, only the first zone of the bladder will expand to the threshold pressure of the zone to securely hug the cargo. Accordingly, the second and third zones of the bladder will remain compressed. As previously discussed, the first zone of the bladder remains expanded until a solenoid valve on the supply device is actuated by an electronic input from an HMI or by a user engaging with a switch on the supply device to reverse the flow of air or water then-present in the bladder back into the supply device. In this way, the apparatus improves the prevention of cargo movement in a vehicle by utilizing an inflatable bladder to secure cargo.

(13) Referring to FIG. 1, an example of an apparatus **100** is illustrated. As described herein, the apparatus is generally a system within a vehicle used to secure cargo. The apparatus **100** includes various elements. It will be understood that in various embodiments it may not be necessary for the apparatus **100** to have all of the elements shown in FIG. 1. The apparatus **100** can have any combination of the various elements shown in FIG. 1. Further, the apparatus **100** can have additional elements to those shown in FIG. 1. In some arrangements, the apparatus **100** may be implemented without one or more of the elements shown in FIG. 1.

(14) Some of the possible elements of the apparatus **100** are shown in FIG. 1 and will be described

along with subsequent figures. It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, the discussion outlines numerous specific details to provide a thorough understanding of the embodiments described herein. Those of skill in the art, however, will understand that the embodiments described herein may be practiced using various combinations of these elements. In either case, the apparatus **100** includes the elements shown in FIG. **1** implemented to perform methods and other functions as disclosed herein relating to improving the prevention of cargo movement in a vehicle.

(15) In one configuration, the apparatus **100** is associated with a cargo area **110**. As used herein, a “cargo area” is a space in a vehicle where cargo may be placed, such as a trunk space, a space behind a seat of a vehicle, or the like. In one or more implementations, the cargo area **110** is a truck bed. While arrangements will be described herein with respect to truck beds, it will be understood that embodiments are not limited to truck beds.

(16) In one approach, the apparatus **100** includes a cover **120** that extends over the cargo area **110**. The cover **120** may be a rigid surface located within a vehicle. For example, the cover may be a roof, a tray, a seat back, or the like. In one or more embodiments, the cover **120** is a tonneau cover. As used herein, a tonneau cover is a cover that spans over a truck bed of a pickup truck.

(17) In one embodiment, the apparatus **100** includes a bladder **150** coupled to the underside of the cover **120**. The bladder **150** may be coupled to the underside of the cover **120** using an adhesive, straps, hooks, connectors, hook and loop fasteners, or the like by integrating the bladder **150** directly into the cover **120** during manufacturing, or the like. That is, in one arrangement, an underside of the cover may serve as a backside of the bladder itself. In any case, when the bladder **150** is coupled to the cover **120** and is in a retracted position, the bladder **150** lies substantially flat against the cover **120** and **150** faces the cargo area **110**. The bladder **150** is, for example, made of a puncture-resistant material, such as rubber. Accordingly, in one arrangement, the bladder **150** touches rigid surfaces in the cargo area **110** when a supply device inflates the bladder **150** without puncturing.

(18) In one configuration, the apparatus **100** includes a supply device **140** connected to the bladder **150** via a connecting member **160**. The supply device **140** may be a device capable of expanding the bladder **150**, such as a compressor, a fill valve, a pump, or the like. The supply device **140** may be located in or external to the vehicle. As such, the location of the supply device **140** is, for example, an inner deck panel of a truck bed, a trunk space, beneath a seat of a vehicle, or any space internal or external to a vehicle that allows the supply device **140** to connect to the bladder **150** with the connecting member **160**.

(19) The supply device **140** fills the bladder **150** with a supply capable of expanding the bladder **150**, such as water, pre-bottled condensed air, or compressed air depending on the type of supply device **140** implemented in the apparatus **100**. For example, if the supply device **140** is a compressor, the supply may be pre-bottled condensed air or compressed air. In one embodiment, if the supply device **140** is a fill valve, the supply may be water. In one configuration, if the supply device **140** is a pump, the pump may supply water or air to the bladder **150**. The connecting member **160** may be any device that allows air or water to flow within it. For example, the connecting member **160** may be a pipe, a channel, a tube, or the like.

(20) In one embodiment, the supply device **140** expands the bladder **150** into contact with cargo **130** present in the cargo area **110**. The supply device **140** expands the bladder **150** when a solenoid valve **180** located on the supply device **140** actuates to a first position. The cargo **130** may be one or more items of any shape, size, and weight that fit in the cargo area **110**. In one approach, the supply device **140** includes a pressure sensor **170** that senses the pressure within the bladder **150**. In one arrangement, the supply device **140** fills the bladder **150** to a threshold pressure. The supply device **140** stops expanding the bladder **150** when the pressure sensor **170** informs the supply device **140** that the bladder **150** has reached the threshold pressure.

(21) The threshold pressure may be set by an input from a user. For example, in one embodiment, a user sets the threshold pressure by entering an input, such as by pressing a button, flipping a switch, or the like, located on the supply device **140** to stop the supply device **140** from expanding the bladder **150** further. Alternatively, in one approach, a user stops the supply device **140** from expanding the bladder **150** further by entering an input on a human machine interface (HMI), such as a touchpad, button, switch, or the like located within the vehicle.

(22) In one approach, a user selects the threshold pressure with an electronic input on an HMI, such as a touchpad, button, switch, knob, or the like located within the vehicle. For example, the user may enter an electronic input on a touchpad, such as a heads-up display, where the user sets the threshold pressure (e.g., 35 psi) for the supply device **140** to fill the bladder **150** to. Alternatively, a user may enter the threshold pressure on an input device located directly on the supply device **140**. In one embodiment, the user selects the threshold pressure from a list of preset threshold pressures. For example, in one arrangement, the user has an option to fill the bladder **150** to a level one threshold pressure, a level two threshold pressure, a level three threshold pressure, etc., where each subsequent level corresponds to a higher threshold pressure for the supply device **140** to fill the bladder **150** to.

(23) In one embodiment, the threshold pressure is based, at least in part, on the dimensions and weight of the cargo **130**. For example, in one approach, a user inputs the approximate dimensions and weight of the cargo **130** present in the cargo area **110** on an HMI located within the vehicle or directly on the supply device **140**. Accordingly, in one embodiment, a user may input details about the cargo **130**, such as the dimensions of the cargo **130**, the weight of the cargo **130**, and the like on a touchpad located within the vehicle. In one approach, a user inputs the exact dimensions of the cargo **130** on an HMI (e.g., 2 feet wide by 2 feet tall by 2 feet long). Alternatively, in one arrangement, a user inputs generalized dimensions of the cargo **130** on an HMI (e.g., small, medium, large, etc.). In one embodiment, a user inputs the exact weight of the cargo **130** on an HMI (e.g., 20 pounds). Alternatively, in one configuration, a user inputs generalized weight information about the cargo **130** on an HMI (e.g., heavy, light, etc.). In one approach, the threshold pressure increases as the dimensions of the cargo decreases. As such, if a user inputs that the cargo **130** in the cargo area **110** is small, the threshold pressure will be high and causes the supply device **140** to expand the bladder **150** more than if the cargo **130** is large because lighter cargo **130** requires higher threshold pressures to remain stationary. In one embodiment, the threshold pressure increases as the weight of the cargo increases. Accordingly, if a user inputs that the cargo **130** in the cargo area **110** is heavy, the threshold pressure will be high and causes the supply device **140** to expand the bladder **150** more than if the cargo **130** is light because heavier cargo **130** can withstand higher pressures without deforming.

(24) In response to the user inputting details about the cargo **130**, the supply device **140** fills the bladder **150** to an appropriate threshold pressure. As such, if a user inputs that the cargo **130** present in the cargo area **110** is small and light, as shown in FIG. 1, the supply device **140** will fill the bladder **150** to a threshold pressure appropriate to tightly hold the cargo **130**. For example, an input indicating that the cargo **130** is small informs the supply device **140** to expand the bladder **150** to a high threshold pressure so that the bladder **150** contacts and secures the cargo **130**. An input indicating that the cargo **130** is light informs the supply device **140** that a high threshold pressure associated with small cargo is inappropriate to secure the cargo **130** and causes the threshold pressure to lower as to not damage the cargo **130**. When the bladder **150** reaches the threshold pressure, the solenoid valve **180** actuates to a neutral position, which prevents the supply (i.e., air or water) from entering or leaving the bladder **150**. The solenoid valve **180**, in one embodiment, automatically moves to the neutral position upon the bladder **150** filling to the threshold pressure.

(25) The bladder **150** remains expanded until the solenoid valve **180** actuates to a second position to reverse the flow of the supply (i.e., air or water) back into the supply device **140**. In one

arrangement, the bladder **150** includes one or more flexible joints within the bladder **150** that extend between a top and bottom surface of the bladder **150**. When the flow of the supply reverses back into the supply device **140**, the one or more flexible joints retracts to bring the bladder **150** to a flat, compressed position. The solenoid valve **180** actuates in response to, for example, an input on the supply device **140**, such as a switch, button, or the like, or an electronic input from an HMI located on or within the vehicle, such as a touchpad, a button, a switch, a knob, a door handle, a tailgate handle, a gear shifter, or the like. For example, a user may actuate the solenoid valve **180** by flipping a switch located on the supply device **140** to a position corresponding to the reversal of supply flow. Alternatively, a user may actuate the solenoid valve **180** by inputting a command on a touchpad, such as a heads-up display, opening a tailgate by pulling on a tailgate handle, parking the vehicle, etc. In any case, upon receiving an input, the solenoid valve **180** actuates to a position to compress the bladder **150** by reversing the flow of supply.

(26) With reference to FIG. 2, one embodiment of the apparatus **100** of FIG. 1 is further illustrated. As shown, the apparatus **100** is associated with the cargo area **110**, which as illustrated in FIG. 2, is a truck bed. In one embodiment, the apparatus **100** further includes the cover **120**, which as illustrated in FIG. 2, is a tonneau cover. In one configuration, the apparatus **100** includes the bladder **150**. In one embodiment, the apparatus **100** includes the supply device **140** connected to the bladder **150** via connecting member **160**.

(27) As illustrated in FIG. 2, the supply device **140** is a compressor, which is comprised of compressor tank **240**, compressor motor **250**, pressure valve **260**, the pressure sensor **170**, and the solenoid valve **180**. In one embodiment, the compressor tank **240** houses air, and the compressor motor **250** functions to compress the air so that it can leave the supply device **140** and fill the bladder **150**. In one configuration, the pressure sensor **170** senses the pressure within the bladder **150**. The pressure valve **260** prevents the pressure within the supply device **140** from getting too high. For example, if the pressure within the supply device **140** reaches a level that may lead to a malfunction of the supply device **140**, such as an explosion, the pressure valve **260** actuates to release air from the compressor tank **240** to the exterior of the vehicle. As previously discussed, the pressure sensor **170** senses the pressure within the bladder **150**. In one configuration, the solenoid valve **180** actuates to allow air to flow into the bladder **150** via air inlet **210** or out of the bladder **150**.

(28) In one embodiment, as illustrated in FIG. 2, the supply device **140** is located between a deck inner panel **220** and a deck side outer panel **230** of the cargo area **110**. As such, the supply device **140** does not take up space in the cargo area **110**, allowing more space for cargo, and the extra space normally present between the deck inner panel **220** and the deck side outer panel **230** are utilized.

(29) In one approach, the supply device **140** sends compressed air supply **270** to the bladder **150** via the connecting member **160**. In one embodiment, the connecting member **160** may include an air release valve **280** at the end of the connecting member **160** that is closest to the bladder **150**. Accordingly, the air release valve **280** releases trapped air in the connecting member **160** to prevent malfunctions and flow blockages to the bladder **150** when appropriate.

(30) With reference to FIG. 3, one embodiment of the cargo area **110** implemented with the apparatus **100** of FIG. 1 is illustrated. In one configuration, the cargo area **110** includes the cover **120** with the bladder **150** located on the underside of the cover **120**. As illustrated in FIG. 3, in one or more embodiments, the bladder **150** expands to hold cargo **130** tightly to the floor of the cargo area **110**. In one approach, the bladder **150** does not expand uniformly over the cargo **130**. For example, the bladder **150** expands further to hold the shorter cargo **130** down, while the bladder **150** does not expand as far to hold the taller cargo **130** down. In any case, the bladder **150** may expand non-uniformly to hold down the cargo **130** present in the cargo area **110**.

(31) With reference to FIG. 4, one embodiment of the apparatus **100** of FIG. 1 is illustrated. FIG. 4 is cross-sectional view of the cargo area **110**. In one configuration, the apparatus **100** includes a

cover, where the cover is trifold tonneau cover **310**. The trifold tonneau cover **310** comprises three different zones (i.e., trifold tonneau cover zone **310(a)**, trifold tonneau cover zone **310(b)**, and trifold tonneau cover zone **310(c)**). In one embodiment, the apparatus **100** includes a bladder coupled to the underside of the trifold tonneau cover **310**, where the bladder comprises bladder zone **320(a)**, bladder zone **320(b)**, and bladder zone **320(c)** that correlate to each section of the trifold tonneau cover **310**. In one configuration, bladder zone **320(a)**, bladder zone **320(b)**, and bladder zone **320(c)** are separate bladders. In one approach, the apparatus **100** includes a supply device **140** that expands each of the bladder zone **320(a)**, the bladder zone **320(b)**, and the bladder zone **320(c)** separately to contact and secure the cargo **130** extending to different distances from the cover **120** to hold the cargo **130** in place. The supply device **140** may send a supply, such as air or water, to each of the bladder zone **320(a)**, the bladder zone **320(b)**, and the bladder zone **320(c)** via the connecting member **160**.

(32) In one configuration, the supply device **140** includes separate pressure sensors that sense the pressure within each zone of the bladder. For example, pressure sensor **340(a)** senses the pressure within the bladder zone **320(a)**, pressure sensor **340(b)** senses the pressure within the bladder zone **320(b)**, and pressure sensor **340(c)** senses the pressure within the bladder zone **320(c)**. The supply device **140** stops the zones from expanding when the zones expand to a threshold pressure for the zones. As previously discussed, the threshold pressure for each zone may be set by a user using an electronic input from an HMI or by inputting the threshold pressure for each zone directly on the supply device **140**. In one embodiment, the threshold pressure for each zone may depend, at least in part, on the dimensions and weight of the cargo **130** located underneath each zone.

(33) Accordingly, in one or more arrangements, at least one bladder zone may not expand if cargo is not present beneath a particular zone. For example, as illustrated in FIG. 4, if the cargo **130** is only present underneath the bladder zone **320(a)**, only the bladder zone **320(a)** will expand into contact with the cargo **130** as it is not necessary for other zones to expand in order to keep the cargo **130** secure. Alternatively, in one or more embodiments, if the cargo **130** is only present underneath the bladder zone **320(a)**, the bladder zones **320(b)** and **320(c)** may expand into contact with the floor of the cargo area **110** to provide extra security to the cargo **130**. In any case, a user may input the threshold pressure for each bladder zone separately to control how each bladder zone expands.

(34) While the embodiment of the apparatus **100** illustrated in FIG. 4 illustrates a cover and bladder with three separate zones, it should be understood that any number of bladder zones may be present in the apparatus **100**, and as such, for every bladder zone, one pressure sensor will correspond to each bladder zone. Further, each bladder zone may expand to a threshold pressure for the zone which may depend, at least in part, on the dimensions and weight of the cargo beneath each bladder zone.

(35) Detailed embodiments are disclosed herein. However, it is to be understood that the disclosed embodiments are intended only as examples. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the aspects herein in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting but rather to provide an understandable description of possible implementations. Various embodiments are shown in FIGS. 1-4, but the embodiments are not limited to the illustrated structure or application.

(36) The terms “a” and “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The phrase “at least one of . . . and . . .” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase “at least one of A, B, and C” includes A only, B

only, C only, or any combination thereof (e.g., AB, AC, BC or ABC).

(37) Aspects herein can be embodied in other forms without departing from the spirit or essential attributes thereof. Accordingly, reference should be made to the following claims, rather than to the foregoing specification, as indicating the scope hereof

Claims

1. An apparatus for securing cargo within a vehicle, comprising: a bladder coupled to an underside of a cover that extends over a cargo area, the bladder facing the cargo area and comprising zones that are separately inflated to hold the cargo in place; and a supply device coupled with the bladder via a connecting member to expand the bladder into contact with the cargo within the cargo area, wherein the supply device expands the zones to separate threshold pressures that depend, at least in part, on attributes of the cargo associated with respective ones of the zones, wherein the attributes include at least dimensions and weight of the cargo that is acquired from inputs on an interface provided by a user.
 2. The apparatus of claim 1, wherein the supply device is at least one of: a compressor, a fill valve, and a pump, and wherein a threshold pressure is increased as the weight of the cargo increases.
 3. The apparatus of claim 1, wherein the cover is a tonneau cover.
 4. The apparatus of claim 1, wherein a location of the cover includes at least one of: a truck bed, a trunk space, and a seatback.
 5. The apparatus of claim 1, wherein the bladder is formed from a rubber material.
 6. The apparatus of claim 1, wherein the supply device expands the bladder with at least one of: pre-bottled condensed air, compressed air, and water.
 7. The apparatus of claim 1, wherein the supply device stops the bladder from expanding when the bladder expands to a threshold pressure, and wherein the threshold pressure depends, at least in part, on dimensions and weight of the cargo.
 8. The apparatus of claim 1, wherein the supply device expands the zones separately to contact and secure the cargo extending to different distances from the cover to hold the cargo in place.
 9. The apparatus of claim 8, wherein the zones correlate with sections of the cover, and wherein the cover is a tonneau cover.
 10. The apparatus of claim 8, wherein the zones include separate pressure sensors, wherein the supply device stops the zones from expanding when the zones expand to a threshold pressure for the zones, and wherein the threshold pressure is increased as the dimensions of the cargo decrease.
-